

Sử dụng độ đo Information Gain, xây dựng cây quyết định cho tập DL sau:

| Day | Outlook  | Temperature | Humidity | Wind   | Play Tennis |
|-----|----------|-------------|----------|--------|-------------|
| D1  | Sunny    | Hot         | High     | Weak   | No          |
| D2  | Sunny    | Hot         | High     | Strong | No          |
| D3  | Overcast | Hot         | High     | Weak   | Yes         |
| D4  | Rain     | Mild        | High     | Weak   | Yes         |
| D5  | Rain     | Cool        | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool        | Normal   | Strong | No          |
| D7  | Overcast | Cool        | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild        | High     | Weak   | No          |
| D9  | Sunny    | Cool        | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild        | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild        | Normal   | Strong | Yes         |
| D12 | Overcast | Mild        | High     | Strong | Yes         |
| D13 | Overcast | Hot         | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild        | High     | Strong | No          |

Attribute: Outlook

Values(Outlook) = Sunny, Overcast, Rain

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$\text{Entropy}(s_{\text{Sunny}}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$$

$$\text{Entropy}(s_{\text{Overcast}}) = -\frac{4}{4}\log_2\frac{4}{4} - 0\log_2\frac{0}{4} = 0$$

$$\text{Entropy}(s_{\text{Rain}}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.971$$

$$\begin{aligned}\text{Gain}(S, \text{Outlook}) &= \text{Entropy}(S) - \frac{5}{14}\text{Entropy}(s_{\text{Sunny}}) - \frac{4}{14}\text{Entropy}(s_{\text{Overcast}}) - \frac{5}{14}\text{Entropy}(s_{\text{Rain}}) \\ &= 0.94 - \frac{5}{14}0.971 - \frac{4}{14}0 - \frac{5}{14}0.971 = 0.2464\end{aligned}$$

Attribute: Temp

Values(Temp) = Hot, Mild, Cool

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$\text{Entropy}(s_{Hot}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1$$

$$\text{Entropy}(s_{Mild}) = -\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6} = 0.9182$$

$$\text{Entropy}(s_{Cool}) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = 0.8113$$

$$\begin{aligned}\text{Gain}(S, \text{Temp}) &= \text{Entropy}(S) - \frac{4}{14}\text{Entropy}(s_{Hot}) - \frac{6}{14}\text{Entropy}(s_{Mild}) - \frac{4}{14}\text{Entropy}(s_{Cool}) \\ &= 0.94 - \frac{4}{14} \cdot 1 - \frac{6}{14} \cdot 0.9182 - \frac{4}{14} \cdot 0.8113 = 0.0289\end{aligned}$$

Attribute: Humidity

Values(Humidity) = High, Normal

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$\text{Entropy}(s_{High}) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.9852$$

$$\text{Entropy}(s_{Normal}) = -\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7} = 0.5916$$

$$\begin{aligned}\text{Gain}(S, \text{Humidity}) &= \text{Entropy}(S) - \frac{7}{14}\text{Entropy}(s_{High}) - \frac{7}{14}\text{Entropy}(s_{Normal}) \\ &= 0.94 - \frac{7}{14} \cdot 0.9852 - \frac{7}{14} \cdot 0.5916 = 0.1516\end{aligned}$$

Attribute: Wind

Values(Wind) = Strong, Weak

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

$$\text{Entropy}(s_{Strong}) = 1$$

$$\text{Entropy}(s_{Weak}) = -\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8} = 0.8113$$

$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= \text{Entropy}(S) - \frac{6}{14}\text{Entropy}(s_{Strong}) - \frac{8}{14}\text{Entropy}(s_{Weak}) \\ &= 0.94 - \frac{6}{14} \cdot 1 - \frac{8}{14} \cdot 0.8113 = 0.0478\end{aligned}$$

Information Gain :

$$\text{Gain}(S, \text{Outlook}) = 0.2464$$

$$\text{Gain}(S, \text{Temp}) = 0.0289$$

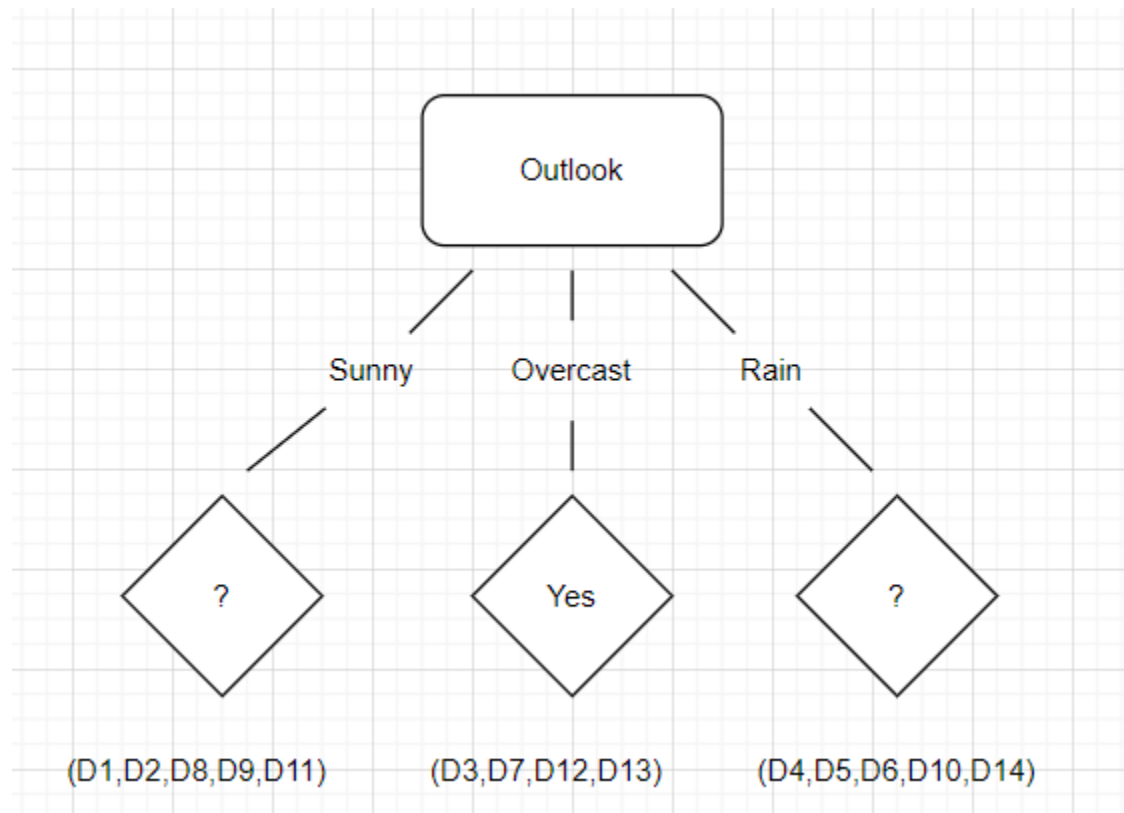
$$\text{Gain}(S, \text{Humandity}) = 0.1516$$

$$\text{Gain}(S, \text{Wind}) = 0.0478$$

Because the Outlook attribute has the highest Information Gain, we choose this attribute to divide the data set with Outlook as the root and the Sunny, Overcast, Rain nodes.

It's clear that with the Overcast value, everyone can play tennis, so we can set up the Overcast - Yes branch.

Next, we calculate the Information Gain between the sunny value and other attributes.



Create Table D1,D2,D8,D9,D11:

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D1  | Hot  | High     | Weak   | No          |
| D2  | Hot  | High     | Strong | No          |
| D8  | Mild | High     | Weak   | No          |
| D9  | Cool | Normal   | Weak   | Yes         |
| D11 | Mild | Normal   | Strong | Yes         |

Attribute: Temp

Values(Temp) = Hot, Mild, Cool

$$\text{Entropy}(s_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$\text{Entropy}(s_{Hot}) = 0$$

$$\text{Entropy}(s_{Mild}) = 1$$

$$\text{Entropy}(s_{Cool}) = 0$$

$$\begin{aligned} \text{Gain}(s_{Sunny}, \text{Temp}) &= \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(s_{Hot}) - \frac{2}{5} \text{Entropy}(s_{Mild}) - \frac{1}{5} \text{Entropy}(s_{Cool}) \\ &= 0.97 - \frac{2}{5} 0 - \frac{2}{5} 1 - \frac{1}{5} 0 = 0.57 \end{aligned}$$

Attribute: Humidity

Values(Humidity) = High, Normal

$$\text{Entropy}(s_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$\text{Entropy}(s_{High}) = 0$$

$$\text{Entropy}(s_{Normal}) = 0$$

$$\begin{aligned} \text{Gain}(s_{Sunny}, \text{Humidity}) &= \text{Entropy}(S) - \frac{3}{5} \text{Entropy}(s_{High}) - \frac{3}{5} \text{Entropy}(s_{Normal}) \\ &= 0.97 - \frac{3}{5} 0 - \frac{3}{5} 0 = 0.97 \end{aligned}$$

Attribute: Wind

Values(Wind) = Strong, Week

$$\text{Entropy}(s_{\text{Sunny}}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$$

$$\text{Entropy}(s_{\text{Strong}}) = 1$$

$$\text{Entropy}(s_{\text{Week}}) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 0.9183$$

$$\begin{aligned}\text{Gain}(s_{\text{Sunny}}, \text{Wind}) &= \text{Entropy}(S) - \frac{2}{5}\text{Entropy}(s_{\text{Strong}}) - \frac{3}{5}\text{Entropy}(s_{\text{Week}}) \\ &= 0.97 - \frac{2}{5} \cdot 1 - \frac{3}{5} \cdot 0.9183 = 0.0192\end{aligned}$$

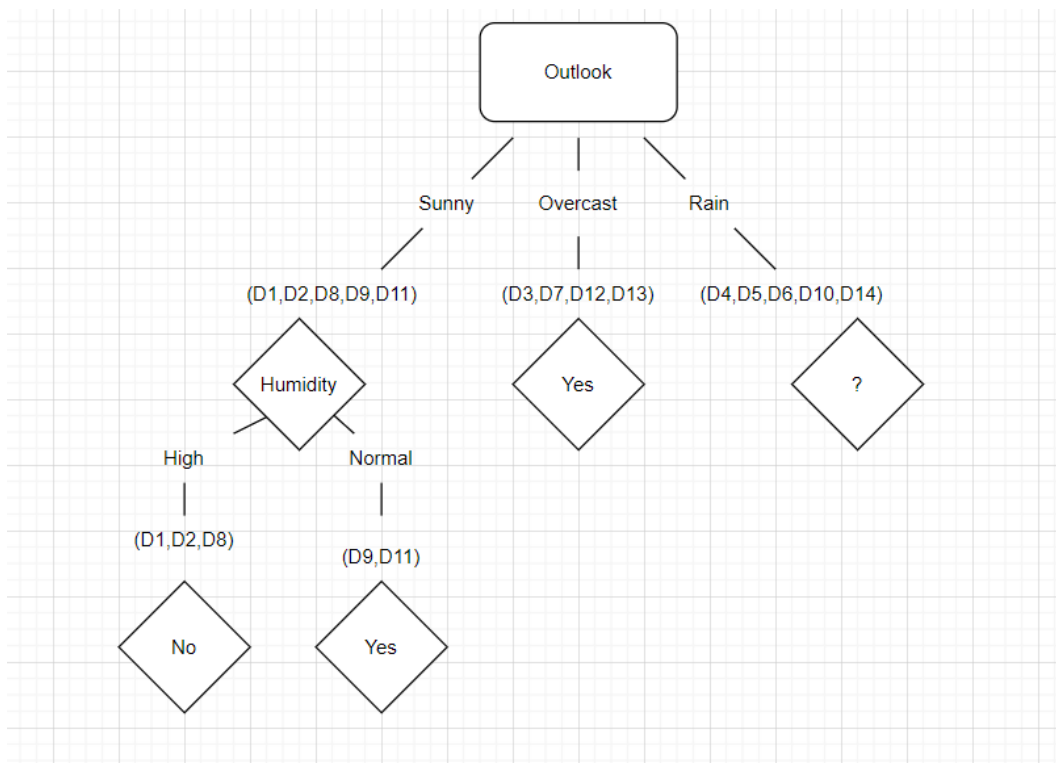
Information Gain :

$$\text{Gain}(s_{\text{Sunny}}, \text{Temp}) = 0.57$$

$$\text{Gain}(s_{\text{Sunny}}, \text{Humidity}) = 0.97$$

$$\text{Gain}(s_{\text{Sunny}}, \text{Wind}) = 0.0192$$

Because the Information Gain characteristic between Sunny and Humidity is the greatest, the branch from Sunny will be Humidity.



Similarly, at the Rain value we have Table D4,D5,D6,D10,D14 :

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D4  | Mild | High     | Weak   | Yes         |
| D5  | Cool | Normal   | Weak   | Yes         |
| D6  | Cool | Normal   | Strong | No          |
| D10 | Mild | Normal   | Weak   | Yes         |
| D14 | Mild | High     | Strong | No          |

Attribute: Temp

Values(Temp) = Hot, Mild, Cool

$$\text{Entropy}(s_{\text{Rain}}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$\text{Entropy}(s_{\text{Hot}}) = 0$$

$$\text{Entropy}(s_{\text{Mild}}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183$$

$$\text{Entropy}(s_{\text{Cool}}) = 1$$

$$\begin{aligned} \text{Gain}(s_{\text{Rain}}, \text{Temp}) &= \text{Entropy}(S) - \frac{0}{5}\text{Entropy}(s_{\text{Hot}}) - \frac{3}{5}\text{Entropy}(s_{\text{Mild}}) - \frac{2}{5}\text{Entropy}(s_{\text{Cool}}) \\ &= 0.97 - \frac{0}{5} \cdot 0 - \frac{3}{5} \cdot 0.9183 - \frac{2}{5} \cdot 1 = 0.0192 \end{aligned}$$

Attribute: Humidity

Values(Humidity) = High, Normal

$$\text{Entropy}(s_{\text{Rain}}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$\text{Entropy}(s_{\text{High}}) = 1$$

$$\text{Entropy}(s_{\text{Normal}}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183$$

$$\begin{aligned} \text{Gain}(s_{\text{Rain}}, \text{Humidity}) &= \text{Entropy}(S) - \frac{2}{5}\text{Entropy}(s_{\text{High}}) - \frac{3}{5}\text{Entropy}(s_{\text{Normal}}) \\ &= 0.97 - \frac{2}{5} \cdot 1 - \frac{3}{5} \cdot 0.9183 = 0.0192 \end{aligned}$$

Attribute: Wind

Values(Wind) = Strong, Week

$$\text{Entropy}(s_{\text{Rain}}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$\text{Entropy}(s_{\text{Strong}}) = 0$$

$$\text{Entropy}(s_{\text{Week}}) = 0$$

$$\begin{aligned} \text{Gain}(s_{\text{Sunny}}, \text{Wind}) &= \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(s_{\text{Strong}}) - \frac{3}{5} \text{Entropy}(s_{\text{Week}}) \\ &= 0.97 - \frac{2}{5} \cdot 0 - \frac{3}{5} \cdot 0 = 0.97 \end{aligned}$$

Information Gain :

$$\text{Gain}(S_{\text{Rain}}, \text{Temp}) = 0.0192$$

$$\text{Gain}(S_{\text{Rain}}, \text{Humandity}) = 0.0192$$

$$\text{Gain}(S_{\text{Rain}}, \text{Wind}) = 0.97$$

We can observe that the Information Gain of Rain and Wind is the biggest, hence the branch associated with Rain will be Wind.

At last, the Decision Tree that follows is obtained:

